Derek P. Horkel

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Hands-on Machine Learning Architect guiding algorithmic data infrastructure and regulated medical AI from model development to deployment. Experienced in scalable ML operations, conducting research and translating into portable production systems built for cloud platforms. Reliable translator between research, production, and engineering teams.

Technical Skills

Programming Languages: Python (statically typed), SQL, C/C++, Terraform, Bash, Node.js, R, Perl, SAS

ML & Data Science: PyTorch, MLFlow, Scikit-learn, XGBoost, Pandas, Polars, NumPy, SciPy, Jupyter, PySpark, OpenCV, NetworkX, Tesseract, Peewee

MLOps & Platforms: ONNX Runtime, TensorRT, Pydantic, ZenML, Kubernetes, Docker, JFrog, Azure DevOps Pipelines

Data & Cloud: AWS (S3, EMR, Athena, Glue, EC2, SageMaker, EKS, ECS, ECR, Batch, DynamoDB, RDS, Redshift, Lambda, CloudFormation, IAM, Rekognition), PostgreSQL, MySQL

Productivity: Mathematica, LaTeX, Linux, Git, Github Copilot, OpenClaw, Cookiecutter

Languages: Spanish (elementary), Mandarin Chinese (elementary)

Employment

Machine Learning Architect, Digital Diagnostics **Jan 2026 – Present**

- Brought regulated medical AI algorithms to trial, including integration, testing, and monitoring
- Bridged communication gap between research and engineering teams, dramatically reducing time to trial for new models
- Delivered production preprocessing and ONNX Runtime inference framework, streamlining deployment and integration
- Implemented cutting edge vision models from papers to deployment, including hybrid CNN-transformer architectures
- Worked with compliance and product stakeholders to create FDA-ready documentation and processes for medical AI
- Introduced best practices for AI-assisted coding, including limitations and guardrails for effective use
- Mentored and led engineers across the ML stack, fostering best practices in software engineering and MLOps

Lead Machine Learning Software Engineer, Digital Diagnostics **Nov 2022 – Dec 2025**

- Designed and built PostgreSQL-based algorithmic data platform to support data ingestion, annotation, training, testing, model versioning, and compliance needs across the entire ML lifecycle, ensuring a unified source of truth
- Introduced CI guardrails that decreased regression-related hotfixes and improved cross-team confidence in releases
- Established best practices for code quality, including introducing static typing and use of cookiecutter templates
- Maintained FDA-cleared AI medical device, upgrading TensorRT/CUDA versions and porting from Windows to Linux
- Managed AWS infrastructure, improving logging and monitoring while optimizing costs for data storage and compute

Senior Machine Learning Engineer, Happy Health **Aug 2021 – Nov 2022**

- Developed models using time series classification, regression, and Markov chain methods
- Ported Python ML models to C to run both in iOS app and on embedded devices
- Coordinated with firmware team to balance sensor data collection with battery demands

Principal Machine Learning Engineer, Sapient Industries **Jun 2019 – Aug 2021**

- As sole machine learning engineer, led end-to-end data collection, labeling, model development, versioning, report generation and production deployment efforts across energy analytics platform
- Iterated and retrained production models, with autonomous tuning for new customers, improving accuracy and reliability
- Managed production, integration test, and development databases hosted in AWS RDS
- Brought AWS accounts into CIS benchmark compliance and implemented automated alerts

Senior Machine Learning Engineer, Vanguard **Dec 2017 – May 2019**

- Deployed, maintained and automated machine learning models and engineered features
- Handled data cleansing, wrangling, and staging for use in models
- Consulted on use of models in marketing campaigns, advising clients, and operations

Postdoctoral Fellow, Temple University **Oct 2016 – Oct 2017**

- Research focused on studying hadron structure using lattice quantum chromodynamics
- Worked with international collaboration coordinating and using computing resources

Education

University of Washington, Seattle, Washington USA

PhD in Physics, 2016; M.S. in Physics, 2012

University of Connecticut, Storrs, Connecticut USA

B.S. in Physics with Honors Math minor; B.S. in Biological Sciences with Ecology and Evolutionary Biology minor, 2011

Patents and Journal Publications

Powered device electrical data modeling and intelligence, US11681345B2 (2023)

Topological susceptibility from twisted mass fermions using spectral projectors and the gradient flow, *Phys. Rev. D97* (2018) 7, 074503

Phase structure with nonzero Θ_{QCD} and twisted mass fermions, *Phys. Rev. D92* (2015) 9, 094514

Impact of electromagnetism on phase structure for Wilson and twisted-mass fermions including isospin breaking, *Phys. Rev. D92* (2015) 7, 074501

Phase diagram of non-degenerate Wilson and twisted mass fermions, *PoS LATTICE2014* (2014) 066

Phase diagram of nondegenerate twisted mass fermions, *Phys. Rev. D90* (2014) 9, 094508